

Water Scarcity Hackathon - Phase 1 - Streamflow Forecasting

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Abstract

This report presents the methodology, approach, and findings from the Capgemini Invent Water Scarcity Hackathon. The project employs a hybrid causal inference and machine learning framework to estimate the causal impact of streamflow drivers. Leveraging CatBoost, a gradient boosting algorithm well-suited for handling categorical variables and missing data, the model incorporates both dynamic and static data sources. Causal analysis is used to identify influential features in time-varying data, while Principal Component Analysis (PCA) is applied to reduce the dimensionality of static variables. The model architecture includes a custom ensemble of gradient boosted trees, optimized using time-series cross-validation to enhance robustness. Additionally, per-fold Quantile Conformal Calibration is implemented to ensure reliable predictive uncertainty and coverage.

1 Introduction

This project was developed as a response to the Streamflow Forecasting and Water Resource Regulation Hackathon, 2025, organized by Capgemini Invent through Codabench. The task was to build models to forecast water flow at stations on river basins in France (Adour-Garonne, Rhône-Mediterranean) and Brazil (Doce River). It involved predicting streamflow four weeks ahead using a combination of historical streamflow, observed weather, and short-term weather forecasts.

Traditional hydrological models, while effective, are often region-specific and computationally intensive. In contrast, machine learning solutions focus on leveraging rich datasets with lagged and contextual features to capture complex temporal and spatial patterns. My approach leverages gradient boosting trees (CatBoost) combined with a chained multi-output regression to model the spatio-temporal nature of streamflow forecasting, explicitly accounting for dependencies across multiple future time steps.

The rest of this project details the data exploration, causal analysis to identify direct drivers of streamflow, feature engineering, and modeling strategies tailored to handle differing dynamics across locations. It concludes with an evaluation of model performance and reflections on lessons learned for building robust, interpretable streamflow forecasting models

2 Data Description

The dataset is also organized in three main splits - train, evaluation and mini challenge sets, to support both temporal and spatio-temporal generalization tasks. The training set is used to train the model on historical data. The evaluation and mini-challenge sets consists of unseen temporal and Spatiotemporal stations to evaluate the model's performance on the public leaderboard.

Dataset Split	Country	Coverage	Num_Stations	Temporal	Spatiotemporal
Training	France, Brazil	1990–2003	39	✓	
Evaluation	France, Brazil	2004–2009	52 - (13 new)	✓	✓
Mini-challenge	Brazil	2011–2015	2		✓

Table 1: Dataset Structure and Coverage Summary

This setup poses two main challenges: temporal robustness across seen stations and generalization to unseen ones. This project addresses these through rigorous feature analysis, thoughtful feature selection, and the incorporation of causal effects to support reliable and responsive forecasting.

3 Methods and Approach

This project follows a structured but iterative process, starting with data preprocessing and feature engineering, followed by data exploration and causal analysis and finally feature selection, modelling experiments and evalutaion. This goal at all steps preceding the modelling phase is to uncover the most relevant features for predictive accuracy, while also ensuring that the model is interpretable and robust. The modelling phase focuses on iteratively testing different architectures and hyperparameters to find the best performing model.

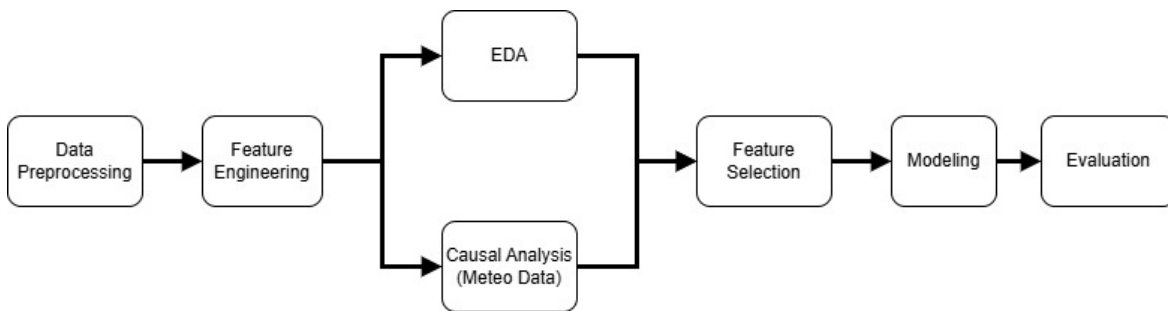


Figure 1: Process Diagram

3.1 Data Preprocessing

Data preprocessing builds on the initial notebook provided in the Competition's Phase 1 GitHub repository, extending it to consolidate and structure information from multiple data sources for downstream modeling.

The workflow enhances the original preprocessing notebook by:

- Integrating local spatial aggregation scales - 50km and 100km buffer scales
- Sampling directly from xarray files for soil and meteorological data. This skips the spatial interpolation step in the original notebook.
- modularizing and organizing all data preprocessing (for train, eval and mini-challenge) into executable scripts orchestrated by a Makefile pipeline for reproducibility and scalability.

3.2 Data Exploration

3.2.1 Contexts, Datasets and Features

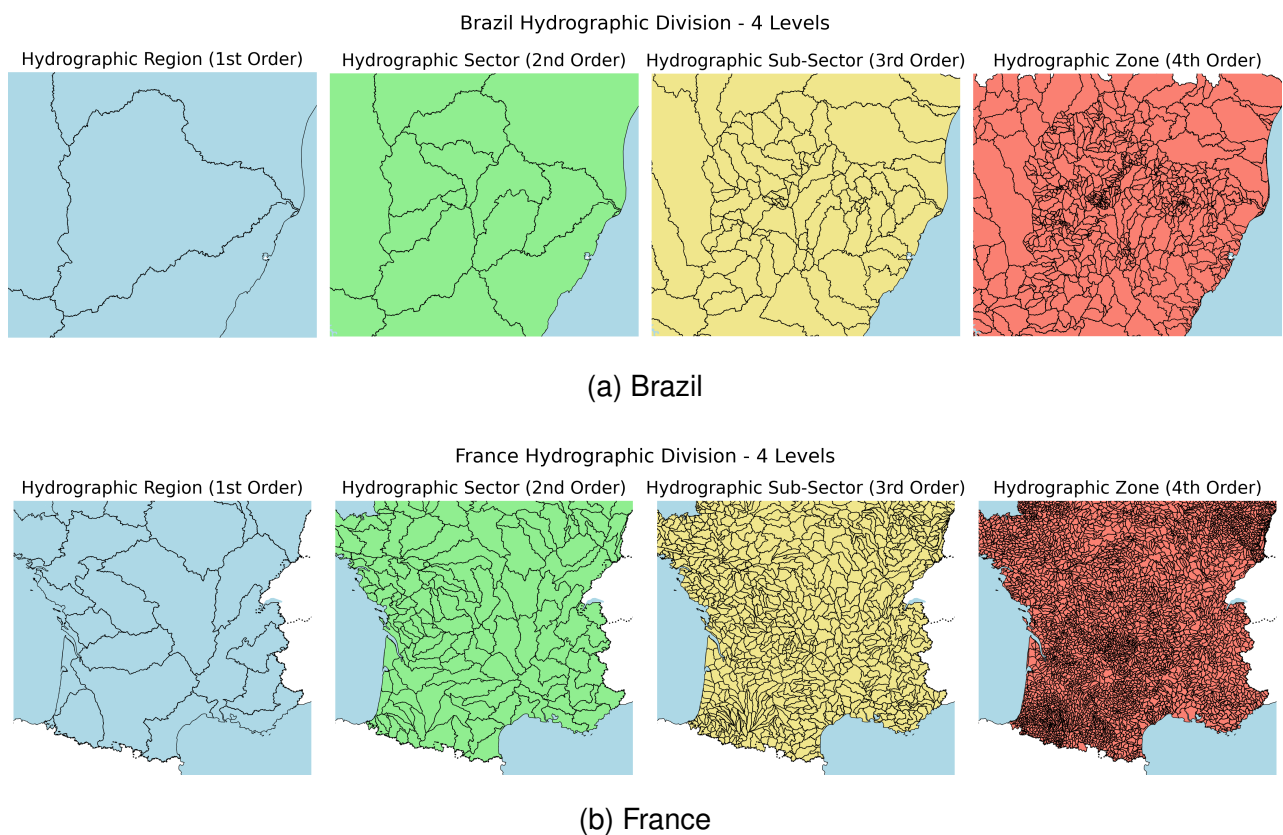


Figure 2: Spatial delineation of hydrological areas in Brazil (top) and France (bottom), across four hierarchical geoscales: region, sector, sub-sector, and zone.

Hydrographic Areas

Hydrographic areas are defined hierarchically—region, sector, sub-sector, and zone. While regions and sectors vary widely in land area across the two countries, sub-sectors and zones exhibit greater spatial consistency, making them ideal for spatial feature aggregation. However, high missingness at these finer scales requires careful imputation, which we address using proximity-based filling from neighboring hydro regions.

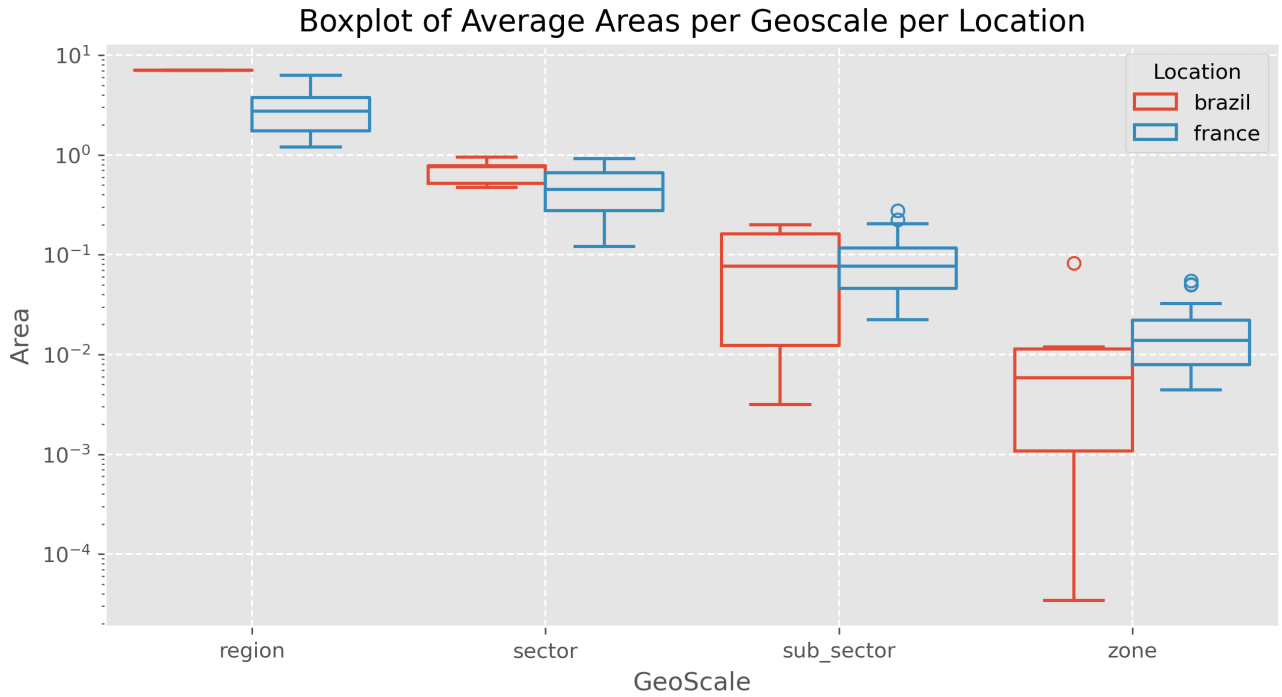


Figure 3: Boxplot comparing areas in the four geoscales in Brazil and France. The maps illustrate spatial variation in size and structure across regions, sectors, sub-sectors, and zones, highlighting differences in spatial consistency between the two countries.

Location Based Features

Location features - (Latitude, Longitude, Altitude) are spatially diverse and capture distinct geographic patterns across regions. These features support location-aware modeling and enhance generalization across spatial contexts. They provide a foundation for learning region-specific hydrological behavior.

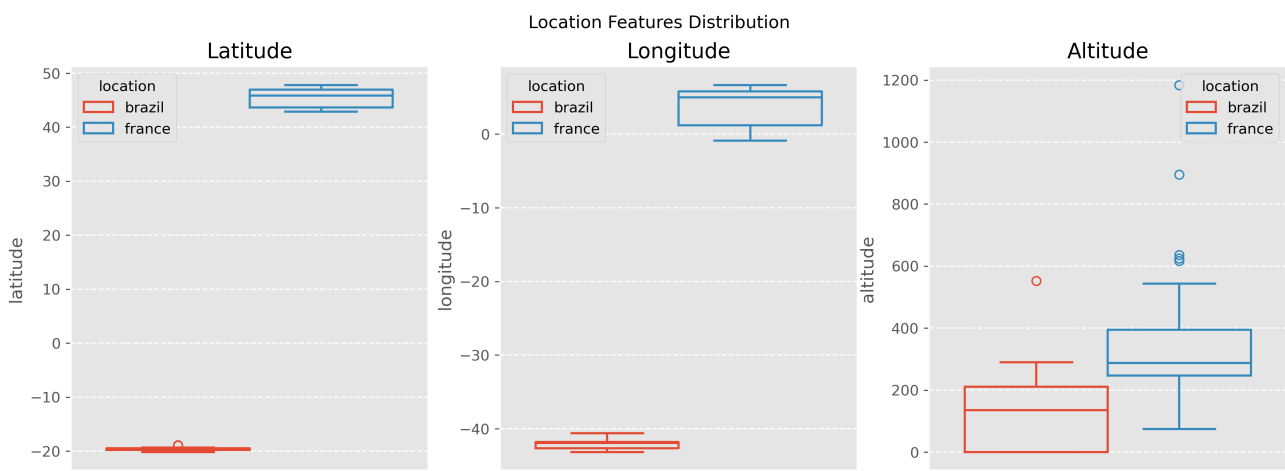
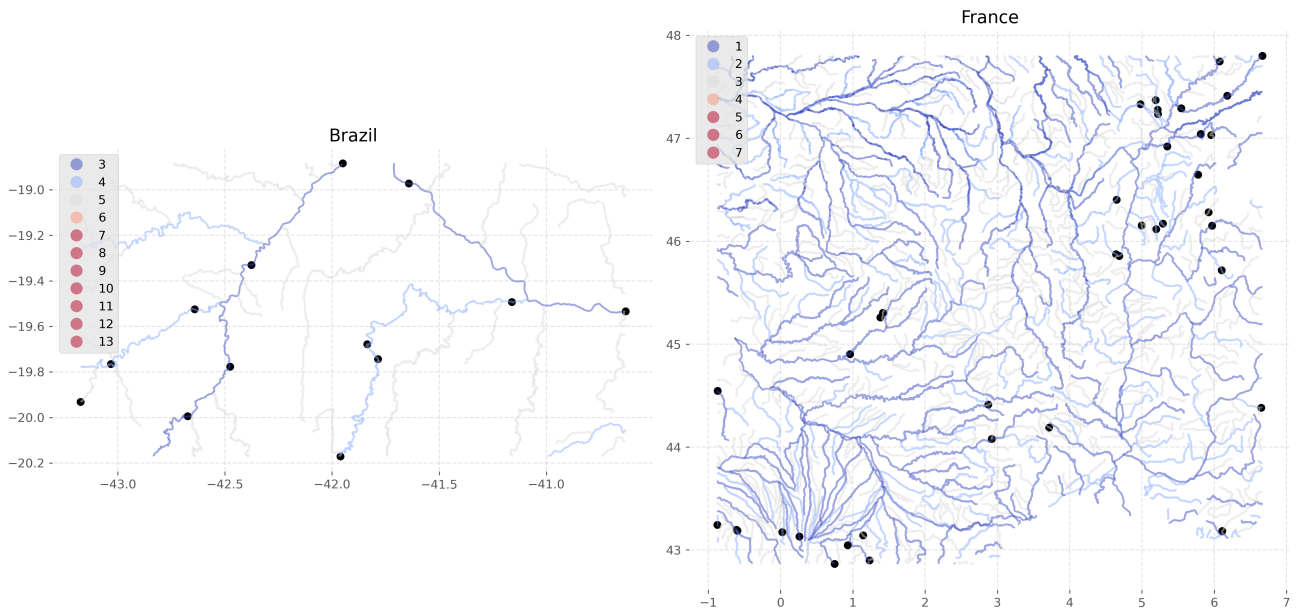


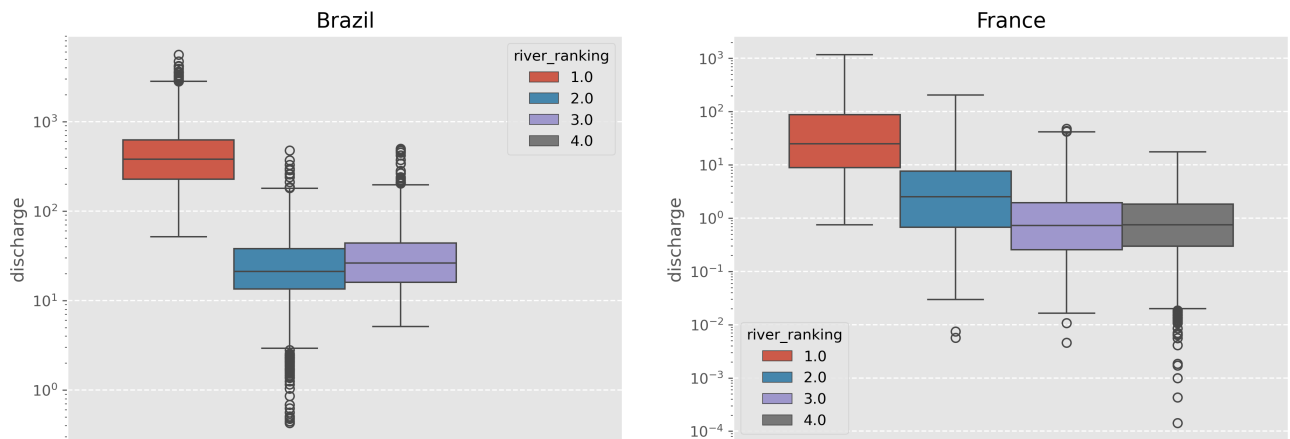
Figure 4: Location-based features across the two countries.

Rivers and Watersheds

Rivers are hierarchically ranked within each watershed, with lower values indicating higher-order rivers. Brazil uses ranks 3–13, while France uses 1–7; we align these by subtracting 2 from Brazil's ranks. Rank 1 rivers in both countries show the highest average discharge. This calibrated river ranking offers a valuable categorical spatial feature for modeling.



(a) River / Stations Spatial Distribution



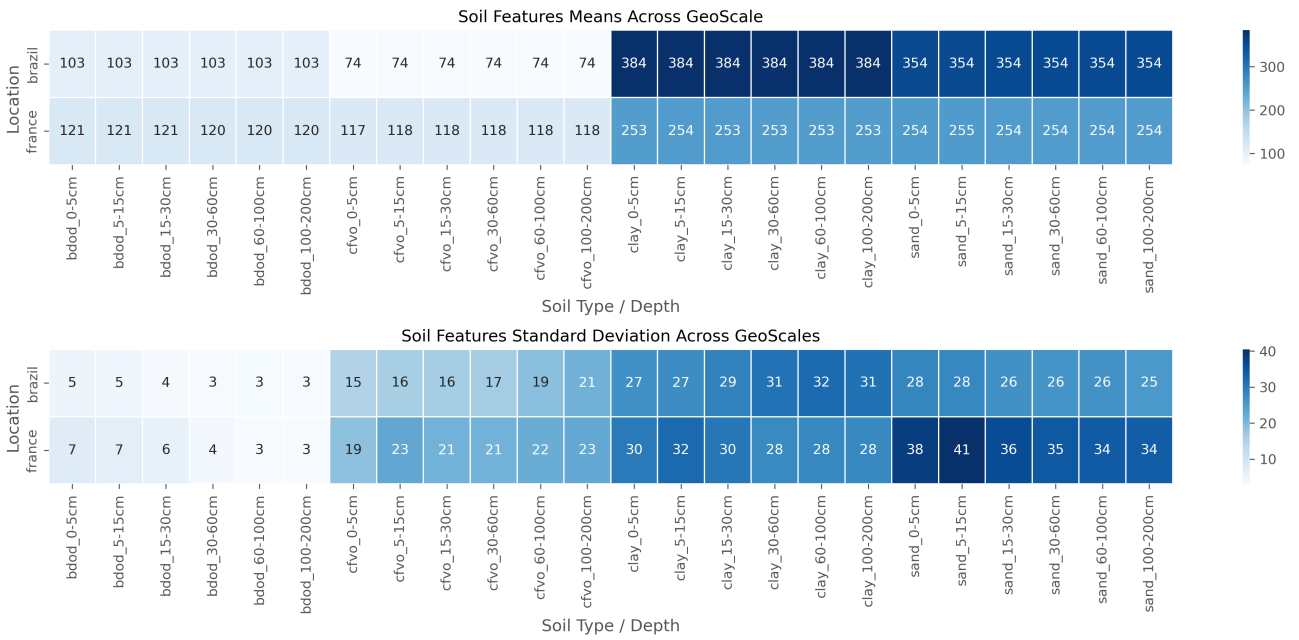
(b) Average Discharge per River Rank

Figure 5: Top: Spatial distribution of top-ranked rivers in each location. Bottom: Average river discharge by rank, highlighting the dominance of higher-ranked rivers across countries

3.2.2 Soil Features

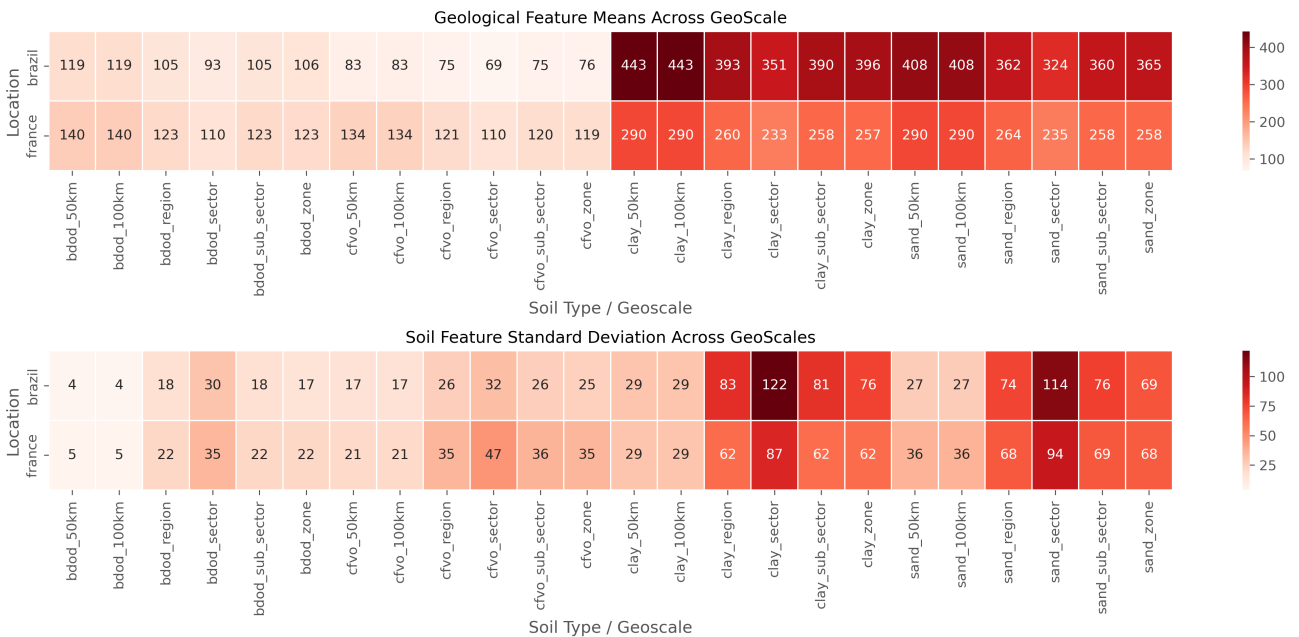
Soil features are static, with no temporal variability, limiting usage to spatial modeling. This project builds on the soil aggregation methodology in the starter preprocessing notebook by extending soil aggregation to incorporate both hydrological scales and buffer zones around each station, enabling a multi-scale spatial understanding of soil properties. For each soil feature layer, both mean values and standard deviations were computed to assess variation.

Soil Feature Profile by Depth



(a) Soil Depth Profile

Soil Feature Profile by Geoscale



(b) Soil Geoscales Profile

Figure 6: Top: Heatmap of soil feature means and standard deviations across depth per soil group. Bottom: Heatmap of soil feature means and standard deviations across geoscales per soil group. Fairly consistent means across depths and geoscales indicate minimal variaton in average soil characteristics.

across depth levels and geospatial units.

The mean values within soil feature groups were generally consistent across depths and geoscales, indicating minimal variation in average soil characteristics. However, standard deviations showed greater variability, particularly in the upper soil layers and at the sector scale. These patterns were consistent across both regions, underscoring the need for careful feature selection to separate discriminative features from noisy ones.

3.2.3 Climate Features

Climate data is aggregated at both the provided hydrological unit scale and within surrounding buffer zones to capture localized spatial variation. Seasonal patterns are analyzed across locations, revealing consistent annual cycles across variables.

Importantly, temperature(t2m) and evapotranspiration (evap) show opposing seasonal trends between northern and southern hemisphere sites due to their geographic positions. To address this, we account for hemispheric differences by enforcing location-specific constraints within the model, ensuring that the model interprets seasonal features within the appropriate spatial context. This prevents confusion from features that exhibit similar meanings but opposite temporal behaviors across regions.

3.2.4 Water Flow Patterns

Per-station analysis of water discharge reveals consistent flow patterns among stations located along the same river systems. Across these stations, discharge exhibits strong annual seasonality with little to no long-term trend. This behavior suggests empirical flow stationarity, indicating that historical discharge data remains representative of current conditions. To leverage this, we compute the annual empirical flow — the historical mean discharge for each week of the year — and include it as a model feature to capture expected seasonal behavior.

For stations not present in the training set, we estimate this feature by taking a weighted mean of the empirical flows from the two geographically nearest stations. This approach provides a reasonable proxy, allowing the model to generalize seasonal discharge patterns to ungauged locations

3.2.5 Engineered Features

Engineering is a crucial step in the modeling process, as it helps create new features that can improve model performance. The following techniques were applied: - OOF Features - KMeans Clustering - Lag features - Rolling features - Seasonal Features - Gaussian 4 / 8 - Temporal features - Categorical features Total Features after feature engineering:

3.3 Feature Selection

We perform feature selection to improve model robustness and reduce overfitting by identifying the most relevant predictors from different feature groups:

- **Soil Features:** To reduce high dimensionality, we select a subset of informative and non-redundant soil variables using statistical and model-based methods. This streamlines the input space while retaining key spatial characteristics.

- **Climate Features:** We prioritize causally relevant climate variables (e.g., precipitation, temperature, evapotranspiration) to promote model robustness and generalizability. This ensures the model focuses on features with direct influence on hydrological responses.

3.3.1 Soil Features

To improve model performance and reduce feature redundancy, we implement a two-step feature selection approach:

1. **Mutual Information Regression (MIR):** We compute MIR scores between each feature and the target variable. MIR is well-suited for identifying relevant soil features due to its ability to capture both linear and non-linear dependencies, which are common in hydrological and environmental data. We retain the top 10% of features with the highest MIR scores, ensuring that the most informative predictors are preserved.
2. **Correlation Filtering:** Within the MIR-selected subset, we apply correlation filtering to remove redundant features. Feature pairs with a Pearson correlation coefficient greater than 0.9 are identified, and the feature with the lower MIR score (or lesser domain relevance) is removed. This reduces multicollinearity and improves model interpretability.

This process yields a compact, non-redundant, and highly predictive feature set that captures both direct and complex interactions with the target variable.

3.3.2 Climate Features

To understand the relationship between climate features and the target variable across different spatial scales, we follow a multi-step approach:

1. **Correlation Analysis:** We first analyze correlations at both local and global levels to identify general patterns and dependencies within the climate data.
2. **Causal Inference:** To move beyond correlation and capture causal effects, we apply causal inference methods at multiple spatial scales, including buffer zones around monitoring sites and broader geographic scales. Specifically, we estimate Average Treatment Effects (ATE) for key climate variables, measuring their direct impact on the target variable while accounting for potential confounders.
3. **Structured Causal Models:** Directed Acyclic Graphs (DAGs) are learned at each spatial scale, consistently identifying temperature and precipitation as important drivers across locations and scales. These findings are further supported by counterfactual analysis, which visualizes potential outcomes under different climate conditions to validate and interpret causal relationships.
4. **Feature Importance:** Focusing on temperature and precipitation features, LightGBM is used to assess feature importance. Selecting the most predictive features within these groups at different scales helps the model better represent key climate drivers.

This hybrid features that combine causal inference insights with machine learning predictions are expected to enhance model robustness by leveraging the strengths of both causal analysis and data-driven methods.

3.4 Modelling FrameWork

The core model is a Gradient Boosting Trees (GBT) regressor trained in a chained forecasting setup, where predictions from previous timesteps (e.g., $\hat{y}_{t-1}, \hat{y}_{t-2}$) are chained into input features for predicting future values $\hat{y}_t$. This autoregressive structure enables multi-step forecasting while capturing temporal dependencies in the data.

To preserve temporal integrity, we use TimeSeriesSplit cross-validation. This ensures the model is always trained on past data and validated on future data. Each fold includes:

- Computation of out-of-fold (OOF) and out-of-year (OOY) empirical flow statistics to avoid leakage.
- Fold specific KMeans clustering to group stations into hydrologically and geographically similar clusters.
- Fold-specific model training.
- Validation residuals computed and conformal calibration factors computed per cluster and per week

At test time, predictions are aggregated by computing:

- The mean forecast.
- Quantiles (0.5, and 0.95) to characterize uncertainty.
- Qantile Conformal Calibration step is applied to improve the coverage and reliability of forecast intervals

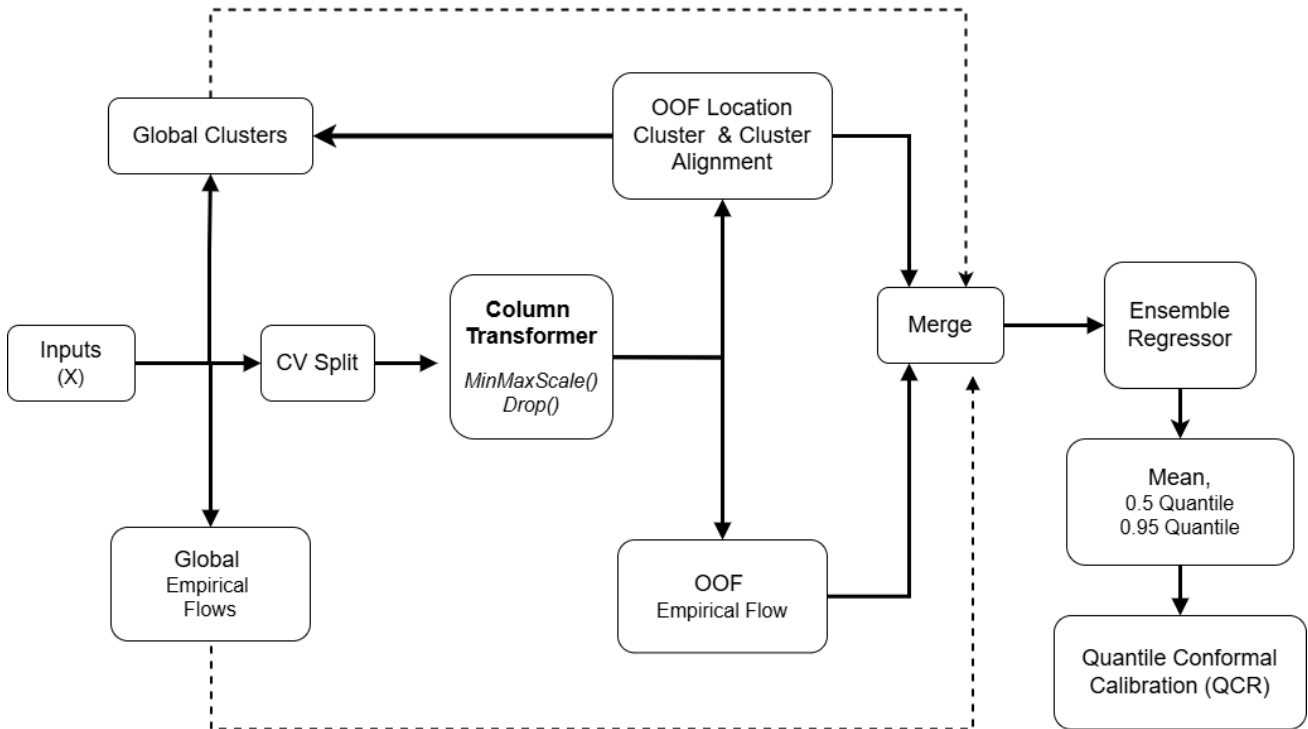


Figure 7: Model Architecture

Algorithm 1 Chained Ensemble Forecasting

```
1: Input: Feature matrix  $X$ , target series  $y$ , number of folds  $C$ , number of ensemble models  $n$ , number of clusters  $k$ , conformal quantile level  $\alpha$ 
2: Output: Forecasts with calibrated quantile intervals  $\hat{y}$ 
3: Compute global KMeans clusters on  $X_{\text{train}}$ 
4: Compute global empirical flow features per station  $F$ 
5: for each fold  $c = 1, \dots, C$  do
6:   Split data into  $X_{\text{train}}^{(k)}, X_{\text{val}}^{(k)}$  using ContiguousTimeSeriesSplit
7:   Compute KMeans clusters on  $X_{\text{train}}^{(k)}$  (per-fold clustering)
8:   Align fold cluster IDs to global cluster ID
9:   Compute OOF and OOF empirical flow features (excluding fold/year)
10:  MinMax scaling of features to  $[0, 1]$ 
11:  for each ensemble model  $i = 1, \dots, n$  do
12:    Train chained GBT regressor on  $X_{\text{train}}^{(k)}$ 
13:    Predict on  $X_{\text{val}}^{(k)}$  and store  $\hat{y}^{(c,i)}$ 
14:  end for
15:  Aggregate fold predictions: mean and quantiles
16:  Compute relative residuals  $r^{(k)} = \frac{\hat{y}^{(k)} - y^{(k)}}{\hat{y}^{(k)} + \epsilon}$ 
17:  Compute  $dt = \text{Quantile}_{1-\alpha}(r^{(k)})$  for each week  $w$ 
18:  Store Conformal factors  $dt$  for each cluster  $c$  and week  $w$ 
19: end for
20: Preprocess test data  $X_{\text{test}}$  using global clusters and empirical flow features
21: Predict on  $X_{\text{test}}$  using the trained ensemble models
22: Compute mean and quantiles of predictions  $\hat{y}_{\text{test}}$ 
23: Smooth conformal factors using weekly temporal blending.
24: Calibrate quantile intervals using smoothed conformal factors.
25: Return ensemble forecasts with calibrated intervals
```

3.4.1 ContiguousTimeSeriesSplit Cross Validation

This validation strategy builds upon the `TimeSeriesSplit` approach from SCIKIT-LEARN, where each fold's training set is a strict superset of the previous one, preserving the temporal ordering of the data. However, our method introduces two key modifications:

- **Contiguous Validation:** Instead of using a fixed-size validation block for each fold, we validate on all remaining data after the training split. This allows the model to be assessed on the full available future horizon, improving robustness and generalization assessment.
- **Minimum Training Size:** A constraint is imposed such that the first training fold must contain at least 50% of the dataset. This ensures sufficient historical context is available for the initial training phase, preventing unstable model estimates due to data sparsity.

The fold structure for a 5-fold setup with years ranging from 1990 to 2004 is defined as follows:

This approach ensures that:

- Each model is trained only on past data relative to the validation set.
- The training window expands over time, simulating a real-world forecasting scenario.

Fold	Training Years	Validation Years
1	1990–1998	1999–2004
2	1990–1999	2000–2004
3	1990–2000	2001–2004
4	1990–2001	2002–2004
5	1990–2002	2003–2004

Table 2: Contiguous expanding time series cross-validation structure.

- The evaluation is performed on all future data from the training endpoint, leading to more realistic performance metrics.

3.4.2 Data-Aware Clustering

To group stations into hydrologically similar regimes, we apply KMeans clustering on the full training data using a fixed number of clusters k . This yields a global clustering structure that informs both model predictions and residual calibration.

To avoid data leakage in time series cross-validation, KMeans is recomputed independently within each training fold. Since the cluster centers may vary between the global and fold-specific KMeans, we align them using the Hungarian algorithm. The alignment minimizes the total Euclidean distance between fold-specific cluster centers C^{fold} and global cluster centers C^{global} .

$$\min_{\pi \in \text{Perm}(k)} \sum_{i=1}^k \|C_i^{\text{fold}} - C_{\pi(i)}^{\text{global}}\|_2 \quad (1)$$

where π is a permutation of cluster indices. The aligned cluster IDs are then used to condition predictions and compute localized residual statistics.

Algorithm 2 Data-Aware Global Clustering and Alignment

- 1: **Input:** Training data X , number of clusters k , number of CV folds F
- 2: Fit global KMeans on X to obtain centers C^{global} and cluster IDs
- 3: **for** each fold $f = 1$ to F **do**
- 4: Extract fold-specific training data $X^{(f)}$
- 5: Fit KMeans on $X^{(f)}$ to get centers $C^{(f)}$
- 6: Compute cost matrix $M_{ij} = \|C_i^{(f)} - C_j^{\text{global}}\|_2$
- 7: Use Hungarian algorithm to solve:

$$\min_{\pi \in \text{Perm}(k)} \sum_{i=1}^k M_{i,\pi(i)}$$

- 8: Map fold-specific cluster IDs using optimal assignment π
 - 9: Assign aligned cluster labels to $X^{(f)}$
 - 10: **end for**
 - 11: **Output:** Data with global-aligned cluster IDs for each fold
-

3.4.3 Empirical Flow Features

Empirical flow statistics are computed as the mean discharge per station and week, excluding the target year to prevent leakage. During training, global empirical flow is calculated and stored. After data splitting, empirical flow is recomputed using only the training fold data.

The inference function `_get_empirical_prior` assigns empirical flow values for each data point as follows:

1. If the station exists in the training priors, assign its empirical flow directly.
2. Otherwise, estimate empirical flow by weighted averaging of the two nearest neighbor stations:
 - Neighbors are selected based on river proximity, river ranking, and geographic location.
 - Euclidean distances on latitude and longitude are computed to find the nearest neighbors.
 - Weights are inverse-distance weighted to combine neighbors' empirical flows smoothly.

This method provides stable, leakage-free empirical priors for both seen and unseen stations during training and inference.

Algorithm 3 Empirical Flow Feature Extraction

Require: Dataframe X with columns `week`, `year`, `station_code`, empirical flow priors `priors`, station metadata `stations_gdf`

Ensure: Dataframe X augmented with empirical flow prior `empirical_flow`

```
1: Assert columns week, year, station_code exist in  $X$ 
2: Copy  $X$  to avoid modifying input
3: for all unique stations  $s$  in  $X$  do
4:   if  $s$  in columns of priors then
5:     Extract empirical flow data for station  $s$ 
6:   else
7:     Find neighbors for  $s$  by:
      • Filtering stations with the same river and similar river ranking and location
      • Calculating Euclidean distance based on latitude and longitude
      • Selecting two nearest neighbors
8:     Compute inverse distance weights for neighbors
9:     Compute weighted average of neighbors' empirical flow by week
10:   end if
11:   Append station empirical flow data to results
12: end for
13: Merge empirical flow results into  $X$  by station_code and week
14: return augmented  $X$ 
```

3.4.4 Relative Quantile Conformal Calibration

To address heteroscedasticity and the varying magnitudes of streamflow across regions and seasons, we propose **Relative Quantile Conformal Calibration (RQCC)** — an extension of

conformal quantile calibration that adjusts prediction intervals using residuals scaled relative to the target value.

In standard quantile conformal prediction, coverage is achieved by computing residuals between predicted quantile bounds and observed values. For a model that outputs lower and upper quantile predictions $\hat{y}_t^{\text{lower}}$ and $\hat{y}_t^{\text{upper}}$, the standard conformal residual is:

$$r_t = \min(\hat{y}_t^{\text{upper}} - y_t, y_t - \hat{y}_t^{\text{lower}}) \quad (2)$$

Using these residuals, the calibrated prediction interval is:

$$[\hat{y}_t^{\text{lower}} - q_{1-\alpha}, \hat{y}_t^{\text{upper}} + q_{1-\alpha}] \quad (3)$$

where $q_{1-\alpha}$ is the $(1 - \alpha)$ -quantile of residuals from a held-out calibration set.

However, in real-world hydrological applications, streamflow values can vary drastically between basins — from small ephemeral streams to major rivers. Applying absolute residuals in such cases results in mis-scaled intervals: overly wide in high-flow regimes and too narrow in low-flow areas. This undermines the interpretability and sharpness of calibrated intervals.

To normalize across such variation in target magnitudes, we define **relative residuals**:

$$r_t^{\text{rel}} = \frac{\min(\hat{y}_t^{\text{upper}} - y_t, y_t - \hat{y}_t^{\text{lower}})}{y_t + \epsilon} \quad (4)$$

where $\epsilon > 0$ is a small constant (e.g., 10^{-6}) added for numerical stability.

Residuals are then grouped by **cluster** c (e.g., hydrological regime) and **timestep** t (e.g., week), and the conformal quantile is computed per group:

$$\delta_{c,t} = \text{Quantile}_{1-\alpha}(\{r_t^{\text{rel}}\}_{i \in \text{cluster } c}) \quad (5)$$

Finally, these relative quantile factors $\delta_{c,t}$ are used to expand the predicted interval proportionally:

$$[\hat{y}_t^{\text{lower}} - \delta_{c,t} \cdot (y_t + \epsilon), \hat{y}_t^{\text{upper}} + \delta_{c,t} \cdot (y_t + \epsilon)] \quad (6)$$

This formulation maintains correct coverage while adapting to local flow characteristics, producing interpretable and consistent intervals across space and time.

4 Experiments and Results

4.1 Experimental Setup

- **Model:** CatBoost with quantile objective (Quantile:alpha=0.1).
- **Validation:** 5-fold contiguous CV with fold expansion, using years as splitting units.
- **Evaluation metrics:** NLL, MAE interval coverage.

4.2 Explainability

4.2.1 Global Feature Importance

Feature importances are computed using the average prediction value change metric:

$$\text{Importance}(x_i) = \mathbb{E}[|f(x) - f(x_{-i})|].$$

4.2.2 SHAP Value Decomposition

We compute Shapley values per station and per cluster to provide local interpretability:

$$f(x) = \phi_0 + \sum_{i=1}^d \phi_i(x).$$

4.2.3 Cluster-wise Interpretation

We group stations by hydrological clusters and visualize dominant features per group, revealing spatial variation in model drivers.

4.2.4 Partial Dependence Analysis

Partial dependence plots are generated for top-ranked features across clusters to examine functional relationships with predictions.

4.3 Robustness

4.3.1 Fold-wise Evaluation

Each fold provides an out-of-fold test set, ensuring realistic generalization. We report:

- RMSE and CRPS per fold.
- Prediction interval coverage at $1 - \alpha$.
- Cluster-wise residual distribution.

4.3.2 Temporal Shift Analysis

To evaluate robustness to temporal shifts, we simulate out-of-year generalization by masking entire years from training.

4.3.3 Input Noise Sensitivity

We perturb key features with Gaussian noise:

$$x'_i = x_i + \mathcal{N}(0, \sigma^2),$$

and observe changes in prediction and coverage metrics.

4.4 Frugality

4.4.1 Data Efficiency

We evaluate model performance using training subsets of increasing size (25%, 50%, 75%, 100%).

4.4.2 Model Complexity Tradeoffs

We benchmark variants with different tree depths and iterations, evaluating performance versus speed and model size.

4.4.3 Inference Efficiency

We report:

- Prediction latency (ms/sample).
- Model memory footprint (MB).

4.4.4 Sparse Station Evaluation

We test the model's ability to generalize to poorly observed stations by holding out target locations and relying on empirical priors.

5 Conclusion

Summarize the key findings and lessons learned from the competition. Discuss potential improvements for future work.

References

List any references or resources used during the competition.

Supplementary Material

.0.1 Data Summary

- **Historical Stream flow:** Streamflow at the station (`.csv`).
- **Hydrographic area:** Hydrographic area geometries (`.shp` / `.geojson` / `.gpkg`).
- **Metereological data:** Temperature, precipitation, soil water volume and evaporation (`.nc`).
- **Geological data:** Soil types (`.nc`)
- **Station Features:** Location, elevation, station_code (`.csv`)

s/n	Description	Value
0	Rows	28509
1	Num Features	387
2	Categorical Features	9
3	Numerical Features	376
4	Location Features	4
5	Unique Locations	2
6	Unique Stations	39
7	Geo Features	336
8	Meteo Features	28
9	Columns with Missing Values	57
10	Total Missing Values	164475
11	Memory Usage	95.26 MB
12	Data Types	bool, datetime64[ns], float64, int64, object
13	Date Range	1990-01-07 to 2004-01-04

Table 3: Training Set Summary Statistics

s/n	Description	Value
0	Rows	6957
1	Num Features	387
2	Categorical Features	10
3	Numerical Features	376
4	Location Features	4
5	Unique Locations	2
6	Unique Stations	54
7	Geo Features	336
8	Meteo Features	28
9	Columns with Missing Values	60
10	Total Missing Values	39155
11	Memory Usage	23.48 MB
12	Data Types	datetime64[ns], float64, int64, object
13	Date Range	2004-01-04 to 2015-03-22

Table 4: Evaluation Set Summary Statistics